Few-Shot Object Detection in Remote-Sensing Images via Label-Consistent Classifier and Gradual Regression

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***Abstract*— With the abomination of time-consuming or even impractical large-scale labeling, few-shot object detection (FSOD) based on natural scenes has attracted extensive attention. However, directly migrating FSOD methods designed for natural images to large-size remote-sensing images (RSIs) still remains a challenge. 1) Labels of novel instances within the base dataset are inconsistently assigned between the base training and the few-shot fine-tuning stage, which confuses the detector and leads to significant performance degradation over novel classes. 2) The region proposal network (RPN) of detectors cannot provide sufficient high-quality proposals for remote-sensing objects with various aspect ratios and irregular shapes, leading to decreased detection performance. To tackle these issues, we specify a novel few-shot object detector for RSIs, to avoid the significant performance degradation caused by inconsistent labeling assignments, as well as efficiently leveraging the novel instances that existed in the base dataset. Furthermore, the proposed detector utilizes a coarse-to-fine regression method with an enhanced feature extractor called Gradual RPN to improve the recall of the RPN. Experiments on a newly constructed few-shot detection benchmark show that our approach improves the mAP of novel classes by up to 8.4% and the average recall of the RPN by up to 12.3%. The source code is available at https://github.com/YanxingLiu/SAE-FSDet.**

***Index Terms*— Few-shot learning, object detection, remote-sensing images (RSIs), transfer learning.**

I. Introduction

In THE past decade, visual object detection has made significant progress, thanks to the exciting representation power of deep convolutional neural networks (CNNs). Nonetheless, existing object detectors heavily rely on large-scale training samples to achieve satisfactory performance, and collecting sufficient labeled data is time-consuming or even unfeasible for many applications. When only a few training samples are available, CNN-based detectors could suffer from overfitting and low detection performance. To address these limitations, few-shot object detection (FSOD) for natural scenes has been proposed and has attracted the attention of a growing number of researchers [1], [2], [3], [4].

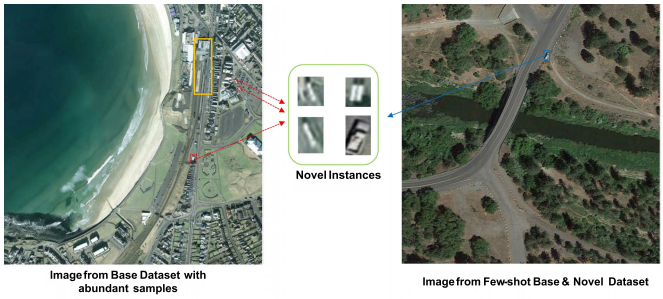


Fig. 1. Inconsistent assignment of unlabeled novel instances between the base training stage and the few-shot fine-tuning stage. In an image from the Base Dataset, there not only exists a *train station* instance belonging to base classes (indicated by the orange box), but also some unlabeled *car* instances belonging to novel classes (indicated by the dashed red boxes). But in the few-shot fine-tuning stage, novel instances are assigned as the foreground to be detected in the Few-shot Base and Novel Datasets.

Many FSOD methods designed for natural images adopt a two-stage training paradigm. The paradigm involves an initial base training stage on an existing base dataset that contains sufficient training samples, to acquire general prior knowledge. Subsequently, the base-trained detector performs fine-tuning on a few-shot dataset that includes target categories. The categories contained in the base dataset are base classes, and the target categories to be detected are named novel classes. The two-stage training paradigm has been validated in various natural scenarios. However, when attempting to migrate these FSOD approaches to RSIs, the following challenges remain.

A. Labels of Novel Instances in the Base Dataset Are Inconsistently Assigned Between Base Training and Few-Shot Fine-Tuning

In the base training stage, novel instances will be assigned as backgrounds, since the base dataset only contains annotations for base classes. By contrast, in the few-shot fine-tuning stage, the instances of novel classes will be assigned as foregrounds because they belong to the target classes that need to be

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detected. As shown in Fig. 1, three *car* instances that need to be detected are assigned as backgrounds in an image from the base dataset, while *car* instances will be assigned as foregrounds in an image from the few-shot dataset used for fine-tuning. As a result, the inconsistent label assignment of novel instances will confuse detectors, resulting in significant performance degradation when detecting novel classes.

The inconsistent label assignment of novel instances not only exists in remote-sensing scenarios, but also exists in natural scenarios [5]. However, for the large image size, remote-sensing images (RSIs) typically contain much more novel instances than natural images. Our experiment shows the ratio of novel instances in the base dataset of the DIOR [6] dataset is 4–5 times greater than that of the PASCAL VOC [7] dataset. This implies that the inconsistent label assignment issue in RSIs is more serious compared to natural scene images.

B. Proposals From the RPN Are of Low Quality

Unlike natural scene objects, remote-sensing objects are often more complex due to their various aspect ratios and irregular shapes. Specifically, due to the difficulty of long-range regression, it is hard for the RPN to regress a remote-sensing object with extreme aspect ratios in a single regression step. In addition, the irregular shape of remote-sensing objects makes it difficult for the RPN to extract the key features of the target. These issues both contribute to the reduced recall of the RPN in the aerial object detection task. Nonetheless, with large-scale and diverse training samples, these issues do not appear to be a problem. With the help of large-scale and diverse data, the RPN can extract distinguishing features of the objects and locate the objects precisely from a variety of diverse training data. Unfortunately, large-scale and diverse training samples do not exist in FSOD. As a result, both the complexity of remote-sensing objects and the sample scarcity in FSOD tasks prevent the RPN from producing sufficient high-quality proposals, thus reducing the performance of detectors. Therefore, it is necessary to design an RPN that is more compatible with the characteristics of remote-sensing objects to enhance its recall in FSOD tasks.

To address the aforementioned issues, we introduce a sample-efficient few-shot object detector named SAE-FSDet. SAE-FSDet not only avoids the performance degradation caused by the inconsistent label assignment of novel instances, but also efficiently leverages the novel instances of the base dataset without any additional time-consuming labeling. More importantly, SAE-FSDet can enhance the recall capabilities of the RPN by a gradual regression method.

First, we propose a label-consistent classifier (LCC) to address the issue of inconsistent label assignment. The LCC is composed of a base classifier and a novel classifier, and the final classification scores are calculated hierarchically by both the base classifier and the novel classifier. In the proposed LCC, unseen novel objects are classified as the *others* category in the base classifier, either in the base training stage or in the fine-tuning stage. The novel instances are classified as their actual classes in the novel classifier. This indicates that the novel instances are assigned consistently in both the base training stage and the few-shot fine-tuning stage. Moreover, with the help of the hierarchical classification scheme, the proposed LCC can effectively leverage the unlabeled novel instances in the base dataset to improve the classification accuracy of newly encountered novel classes.

In what follows, we designed a delicate region proposal network (RPN) named gradual region proposal network (Gradual RPN) that can generate high-quality proposals to enhance the recall of the detector. Instead of directly generating proposals from predefined anchors, Gradual RPN takes a coarse-to-fine regression method to gradually regress remote-sensing objects with various aspect ratios. It also utilizes a deformable convolutional network (DCN) [8] to improve the ability to extract the key features for irregularly shaped remote-sensing objects.

As the FSOD performance is sensitive to sample selection, for a given method, it could vary slightly depending on the sample selection. For a fair comparison with other few-shot object detectors [3], [4], [9], [10], [11], we constructed a detection benchmark on the DIOR [6] dataset and the NWPU VHR-10 [12] dataset. Experiments show the introduction of Gradual RPN notably improves the recall of the RPN by 12.3%. Additionally, the use of LCC can further improve the detection mAP of novel classes in few-shot scenarios by 8.4%.

The main contributions of this article can be summarized as follows.

1) A label-consistent classifier called LCC is proposed. It can improve the detection performance on multiple datasets by solving the inconsistent assignment of unlabeled novel objects. To the best of our knowledge, this is the first attempt to address the issue of the inconsistent assignment problem prevalent in the remote-sensing community.
2) We propose the Gradual RPN, which can produce sufficient higher-quality proposals for FSOD tasks in aerial images.
3) We construct an FSOD benchmark on the DIOR and NWPU VHR-10 datasets and conduct extensive experiments on the benchmark. Hopefully, the benchmark will contribute to the advancement of FSOD in the field of remote sensing.

The following is the organization of the remaining parts of this article. Section II briefly introduces the related work. Section III provides preliminary knowledge about FSOD and describes our proposed SAE-FSDet method in detail. Section IV reports and analyzes the comparative results on two remote-sensing FSOD datasets. Finally, we conclude this article in Section V.

II. Related Work

A. Generic Object Detection

Deep-learning-based object detectors can be roughly into two types: two-stage detectors and one-stage detectors. Two-stage detectors such as Faster R-CNN [13] first generate a set of region proposals by a region proposal network named RPN, representing the potential foreground regions. Then,

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a fixed shape of features will be extracted by ROI pooling for each proposal and the instance-level features will be classified and regressed to get the final predicted bounding boxes with corresponding confidence scores by a transformation module, a classifier, and a regressor. One-stage detectors such as YOLO series [14], [15], [16] and some anchor-free approaches [17], [18], [19] predict bounding boxes and class scores simultaneously based on a set of predefined anchor boxes, making it extremely fast in inference.

Although object detection for natural scene images has achieved rapid development [6], due to the unique bird-view of RSIs and the domain gap between natural images and RSIs [20], [21], directly applying detectors designed for natural scene images to RSIs will face the following challenges.

1) In RSIs, foreground objects occupy a smaller portion of the overall image. Moreover, the background of RSIs tends to be more complex, which poses challenges in distinguishing objects from the surrounding context.
2) Furthermore, objects in RSIs tend to exhibit larger variations in both shape and scale, which increases the difficulty of detecting objects.
3) The RSIs are taken from an overhead view, thus the angle of objects is not fixed and the facing direction is variable. This makes it difficult to represent remote-sensing objects in conventional horizontal boxes.

To address the above challenges, numerous research [22], [23], [24], [25], [26], [27], [28], [29], [30], [31], [32], [33] have been conducted. To handle the problem of rotation invariance for feature learning, Mei et al. [22] proposed a novel cyclic polar coordinate convolutional layer. Considering that aerial images tend to have large sensory areas and contain multiscale objects, Deng et al. [23] proposed a multiscale object proposal network (MS-OPN) which consisted of three proposal branches to predict multiscale proposals. In addition, the unique bird views of RSIs result in arbitrary orientations of geospatial objects. To achieve rotation-invariant detection, many researchers [24] work on detecting objects with horizontal bounding boxes. For example, Cheng et al. [24] and Li et al. [25] proposed a rotation-invariant CNN (RICNN), which contains a novel rotation-invariant layer. To detect arbitrary orientation objects, Li et al. [25] proposed a novel RPN with additional multiangle anchors. Since the horizontal bounding boxes are unable to finely represent the shape of aerial objects, plenty of works [26], [27] aim at detecting novel objects with rotated bounding boxes. For instance, Yang et al. [26] introduced an improved smooth L1 loss to deal with the boundary problem for the rotating bounding box. By gliding the vertex, Xu et al. [27] proposed a new regression method to detect rotated boxes. In addition, to detect multiscale objects in RSIs, Liu et al. [28] use a bidirectional feature pyramid network to fuse the cross-scale features. Lian et al. [29] proposed a novel contextual background attack framework to fool aerial detectors. Besides, there also exists extensive research [30], [31], [32], [33] to detect aerial objects.

Detectors, as mentioned above, have achieved excellent performance but require a sufficient number of bounding box annotations and are prone to overfitting when the labeled data becomes scarce. However, collecting sufficient well-labeled data is laborious even impractical, especially for specific tasks or categories.

B. Few-Shot Object Detection

Since massive labeled datasets are not easily available, there are intensive works on FSOD in natural scene images. Current FSOD methods can be divided into meta-learning methods [4], [34], [35], [36] and transfer-learning methods [1], [3], [37].

For meta-learning methods, Kang et al. [34] proposed a detector based on YOLOv2 [15] named few-shot object detection via feature reweighting (FSRW) aiming to re-weight the importance of feature maps with channel attention and then adapted the re-weighted features to improve the detection performance of few-shot novel classes. Instead of re-weighting the feature map used in FSRW, meta R-CNN [35] only re-weight the ROI features and gets better detection performance. Furthermore, FsDetView [36] proposed a novel feature aggregation scheme, which utilizes features from base classes to get more accurate detection results on novel classes. From the perspective of attention, the RPN, FSOD [4] leverages support information to filter out most backgrounds and those in nonmatching categories. VFA [11] first introduced variational feature learning into FSOD to get more robust features. To generate more specific and representative prototypes for each query image, Lu et al. [10] proposed an Information-Coupled Prototype Elaboration (ICPE) method and achieves better results in FSOD on natural images.

Transfer-learning methods aim at fine-tuning the prior knowledge learned from a base dataset with abundant annotated instances to the novel categories with only a few data. LSTD [1] is one of the early works of transfer-learning methods, which aims to leverage rich source-domain knowledge to construct an effective target-domain detector. It is worth noting that LSTD suggests that the two-stage detectors with a coarse-to-fine approach are more effective in extracting common features of foreground targets compared to the single-stage detectors. This is particularly advantageous in scenarios with FSOD as it helps to mitigate the risk of overfitting. Then, TFA [3] proposed a simple, yet effective transfer-learning method, which only fine-tunes the box classifier with instance-level normalization and achieves comparable performance with meta-learning methods. Considering the inconsistent optimization target of the RPN [13] and RCNN [13], DeFRCN [37] designed the gradient decoupled layer to scale or stop the gradient from the RPN and RCNN, leading to a significant performance improvement of novel classes. Meanwhile, by introducing a contrastive proposal encoding loss, FSCE [9] significantly improved the performance of FSOD. Then, to better utilize the prior knowledge from pretrained models, imTED [38] migrates pretrained transformer encoder–decoders to a detector and builds a fully pretrained detector.

Recently, FSOD tasks in RSIs have attracted growing attention. Li et al. [39] proposed a meta-learning method with multiscale detection for the remote-sensing FSOD task. Cheng et al. [40] designed Prototype-CNN, which mainly consisted of a prototype learning network (PLN), a prototype-guided RPN (P-G RPN), and a detection head for detecting

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novel objects. Similarly, Le Jeune et al. [41] propose a prototypical network based on representation learning. To mitigate the severe data imbalance between the novel class and base class, Huang et al. [42] proposed a novel balanced fine-tuning strategy and introduced a shared attention module to leverage the rich ground information in RSIs. PAMS-Det [43] employs the involution operator and shape bias to enhance the classification branch and introduce a multiscale path-aggregation module to improve the regression branch. Wolf et al. [44] propose a double-head few-shot detector named DH-FSDet and introduce an annotation sampling and preprocessing strategy. To address the limited scale diversity in FSOD, MSOCL [45] introduces a multiscale object contrastive learning framework and integrates a novel contrastive multiscale proposal (CMSP) into it. In addition, Zhang et al. [46] introduced Generalized FSOD into remote sensing and proposed a novel efficient transfer-learning framework.

Although previous studies have yielded promising results in FSOD, these studies failed to address the problem of inconsistent assignment of unlabeled novel instances in FSOD. Additionally, the recall of the RPN for complex remote-sensing objects is still unsatisfactory. To solve the above problems, we proposed Gradual RPN with a step-wise regression method to obtain more high-quality candidate boxes in remote-sensing few-shot scenarios and we proposed an LCC that assigns the unlabeled novel instances consistently.

III. Methodology

A. Background

1) Task Definition: In this study, the dataset consists of two types of classes: base classes C_b and novel classes C_n. The base classes are fully annotated and have plenty of annotated object instances, whereas the novel classes have only limited annotations. To avoid overfitting, few-shot detectors usually adopt a two-step training schedule. First, the whole network is trained on a comprehensive training set D_b consisting of base classes to acquire general knowledge about the target task. Second, the detector is fine-tuned on a sample-balanced dataset D_f, comprising both base and novel classes and in the fine-tuning process, the backbone of the network is kept frozen. In D_f, each class comprises K annotated objects. When there are N novel classes, it is referred to as the N-way K-shot FSOD setting.

2) Base Detector: We use TFA [3] as our basic detector, which is modified by introducing a cosine similarity-based classifier to Faster-RCNN [13]. In TFA, only the box predictor is adjusted using D_f, while other components are left frozen. However, in the remote-sensing domain, we find that this adjustment strategy limits knowledge transfer to novel classes and results in poor few-shot novel class performance. Thus, in the fine-tuning stage, we only froze the backbone network and unfreeze the RPN and RCNN to capture high-level features of the novel classes. Fig. 2(a) illustrates the training framework of our proposed method and the joint loss of the whole detector is

$$\mathcal{L} = \mathcal{L}_{\text{rcnn}} + \alpha \mathcal{L}_{\text{rpn}} \tag{1}$$

where $\mathcal{L}_{\text{rcnn}}$ is a loss function from the LCC module and $\mathcal{L}_{\text{rpn}}$ is applied to the Gradual RPN module. α is a hyperparameter to balance the RPN loss as well as the RCNN loss, and we set it to 0.7 in this study.

B. Label-Consistent Classifier

In many previous FSOD methods based on transfer learning [3], [9], [37], classifiers for novel classes are added directly to the end of the base classifier as Fig. 3 shows. However, due to the inconsistent label assignment of novel instances in the base dataset, this classification paradigm will confuse detectors and result in significant performance degradation.

Therefore, we proposed a simple, yet effective LCC. It employs a hierarchical classification paradigm to avoid the inconsistent label assignment of novel instances in the base dataset. In addition, the LCC can further utilize the unlabeled novel instances in the base dataset to enhance the detection performance for novel classes.

As illustrated in Fig. 2(b), our proposed LCC module comprises a base classifier and a novel classifier. The base classifier is trained on the base dataset and the unseen novel objects are all categorized into the *others* category in the base classifier. During the few-shot fine-tuning stage, an extra novel classifier is introduced to the detector that is responsible for classifying the unseen novel objects into their actual classes. The classification probabilities of LCC are calculated from the prediction scores of the base classifier and that of the novel classifier. Specifically, the final classification probabilities for both base and novel classes are calculated as follows:

$$P_i = \begin{cases} \dfrac{\exp(b_i)}{\sum_{j=0}^{|C_b|}\exp(b_j)}, & C_i \in C_b \\ \dfrac{\exp(b_{|C_b|})}{\sum_{j=0}^{|C_b|}\exp(b_j)} \times \dfrac{\exp(n_i)}{\sum_{j=0}^{|C_n|}\exp(n_j)}, & C_i \in C_n. \end{cases} \tag{2}$$

In the above equation, b_i and n_i represent the prediction scores of the ith base class and the ith novel class.

As shown in (2) and Fig. 2(b), for base classes, the predicted probabilities are obtained by simply applying a Softmax function to the prediction scores of all base classes. It should be noted that the probability of the last class calculated from the base classifier is defined as P_{others}

$$P_{|C_b|} = P_{\text{others}} = \frac{\exp(b_{|C_b|})}{\sum_{j=0}^{|C_b|}\exp(b_j)} \tag{3}$$

which includes the probabilities of all unseen novel classes and backgrounds. In (3), $|C_b|$ is the last class of base classifier and b_j represents the jth prediction score.

For novel classes, they simultaneously exist in the base classifier as the others category, while existing as their actual classes in the novel classifier. Thus, both P_{others} in the base classifier and predicted probabilities in the novel classifier should be considered in the calculation of the final probabilities for novel classes. Because a novel class could be regarded as a subclass of *others* class, we argue that multiplying the probability of the *others* category (the last class of the base classifier) with the predicted probabilities in the novel classifier could be a simple yet effective method, to get the

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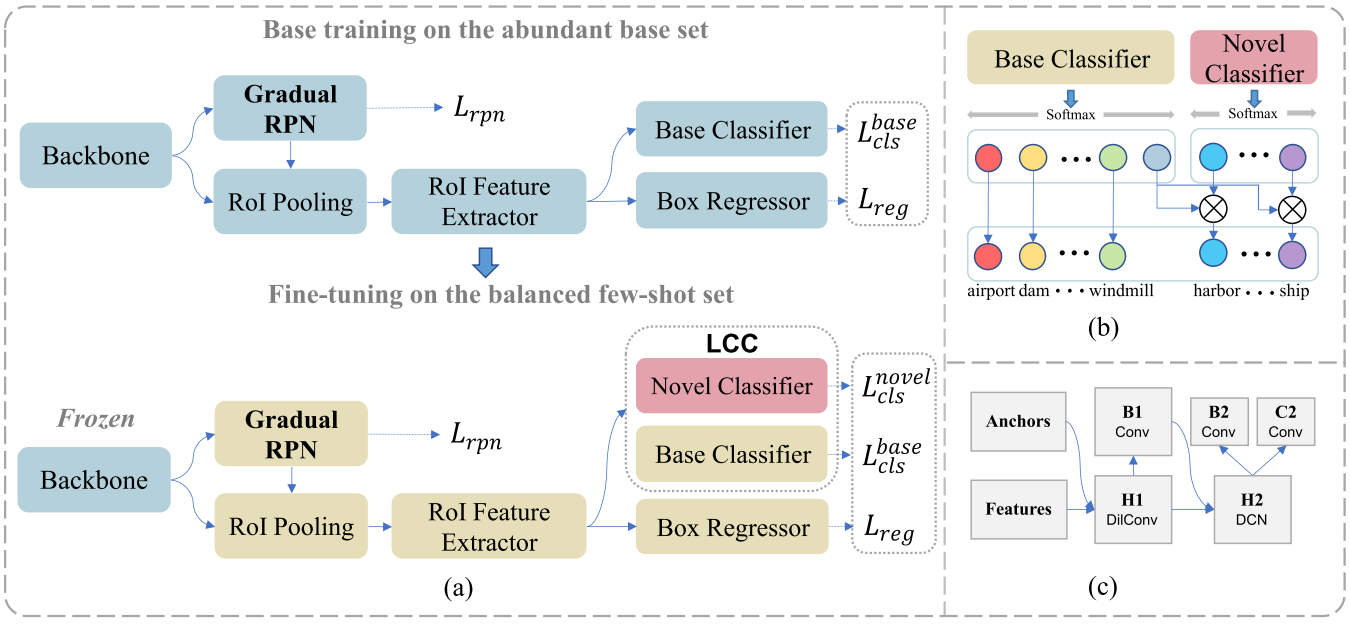


Fig. 2. (a) Overall training framework of our proposed SAE-FSDet. Based on the transfer-learning framework, the entire detector is first trained on the abundant base set and then fine-tuned on the balanced few-shot set with a frozen backbone network. (b) In the fine-tuning stage, the base classifier is replaced with our proposed LCC to avoid the inconsistent assignments of unlabeled novel objects. The novel instances are classified as *others* category (darker blue one in the figure) in the base classifier and are categorized into their actual classes in the novel classifier. The final prediction scores of novel instances are calculated by multiplying the probability of the *others* category with their specific probabilities in the novel classifier. (c) Proposal boxes are regressed gradually from predefined anchors to precise boxes in two regression stages. "H," "C," and "B" in (c) denote feature extraction Head, Classifier, and Boxes predictor, respectively. The features from the backbone are first extracted by a dilated convolution layer with a large receptive field, and the coarse boxes are regressed from predefined anchors. A DCN is then utilized for extracting key features of irregularly shaped remote-sensing objects, and the precise boxes will be regressed from the coarse boxes.

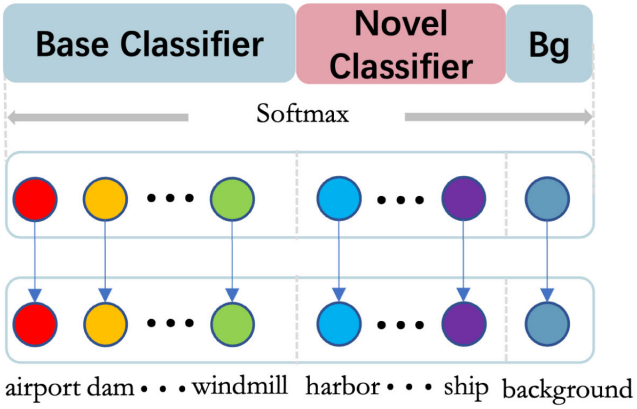


Fig. 3. Classifier in previous FSOD methods based on transfer learning. In the base training stage, unlabeled novel instances, such as ships, are assigned as backgrounds. However, in the fine-tuning stage, these ships are assigned as novel classes to be detected. The inconsistent label assignment will result in a large number of instances belonging to novel classes being incorrectly identified as backgrounds.

final predicted probabilities for novel classes. This method of calculation ensures that the sum of probabilities equals 1 for both the base classes and the novel classes. Furthermore, it matches our intuitive understanding that novel classes are assigned to the others category in the base classifier and to their actual classes in the novel classifier.

In the proposed LCC, during the base training stage, all instances that belong to novel classes are all assigned as others category in the base classifier. In the fine-tuning stage, these novel instances still exist as *others* in the base classifier, indicating that their labels are assigned consistently and will not confuse the detector. The novel classifier is responsible for recognizing instances in *others* category into their specific classes. Thus, we argue that this classification paradigm recognizes base and novel instances in a label-consistent approach.

In addition, as (2) shows, the final classification probabilities of novel classes are not only related to the novel classifier, but also P_{others} in the base classifier. Therefore, assigning novel instances that lack labels to the *others* category in large-scale base training can increase P_{others} when detecting novel objects and further enhance our approach's ability to detect few-shot novel classes. In conclusion, by our proposed LCC, we not only solve the problem of inconsistent assigning of unlabeled novel instances, but also utilize them to improve the detection performance of novel classes.

At last, we will introduce the loss function of LCC. The loss function of our proposed LCC module is composed of three parts

$$\mathcal{L}_{\text{rcnn}} = \mathcal{L}_{\text{cls}}^{\text{base}} + \mathcal{L}_{\text{cls}}^{\text{novel}} + \mathcal{L}_{\text{reg}}. \tag{4}$$

$\mathcal{L}_{\text{cls}}^{\text{base}}$ is a cross-entropy loss used for the base classifier and $\mathcal{L}_{\text{reg}}$ is a smooth $L1$ loss for the box regressor of RCNN. Additionally, as (5) shows, we use a modified multiclass focal loss [47] as the loss function of a novel classifier to mitigate the positive and negative sample imbalance in few-shot scenarios. γ in (5) is a hyperparameter to mitigate the imbalance of positive and negative samples. y_i and p_i are the ith ground truth and prediction, respectively,

$$\mathcal{L}_{\text{cls}}^{\text{novel}} = \sum_{i=0}^{|C_n|} -(y_i - p_i)^{\gamma} \log(p_i). \tag{5}$$

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It should be noted that the original multiclass focal loss [47] employed a sigmoid activation function which is inadequate for this hierarchical classification. Therefore, we opted to use a Softmax activation function for the novel classifier and designated the last class of the classifier as the background class.

C. Step-by-Step Regression: Gradual RPN

The quality of RPN proposals is crucial in Faster R-CNN [13] as inadequate RPN recall will result in reduced detector recall. However, in few-shot scenarios, it is difficult to regress remote-sensing objects with various aspect ratios in just one step and extract adequate features from irregularly shaped remote-sensing objects using a single convolutional layer. Consequently, the inaccurate proposals and the inadequate feature extraction will lead to a low recall of novel classes in remote-sensing FSOD.

To solve the above problems, inspired by Cai and Vasconcelos [48] and Vu et al. [49], we designed a step-by-step regressor named Gradual RPN, which generates more high-quality proposals with limited labeled samples. In Gradual RPN, to overcome the difficulty of long-range regression for objects with extreme aspect ratios, we adopt a coarse-to-fine two-step regression method. We then use a variety of convolutional layers to enhance the feature extraction from the irregularly shaped remote-sensing objects. The stepwise regression structure and diverse feature extraction layer of our Gradual RPN enable it to generate a larger number of high-quality proposals even with few-labeled samples, thereby improving the detection performance of the detector.

As Fig. 2(c) illustrates, the features generated from the backbone network are fed into the first stage of Gradual RPN to obtain the coarse boxes from the predefined anchors. However, hand-designed anchors in Faster R-CNN [13] cannot adapt to the diverse shapes of objects in RSIs (e.g., extremely thin objects like *harbor*). Fortunately, thanks to Gradual RPN's two-step regression setup, there is no need to manually specify the aspect ratios of anchors, and we simply use a square with a scale of 8 as the size of our predefined anchors. The model can generate diverse shapes of coarse boxes automatically by learning in a diverse base dataset. To expand the receptive field, a dilated convolution layer [50] is utilized to extract coarse features in the first regression stage. Meanwhile, to reduce computational complexity, we do not predict the foreground probability of objects and only predict the location of coarse boxes in the first regression stage. By the first stage of regression, coarse boxes are regressed from predefined square candidate boxes. The aspect ratios and scales of coarse boxes are approximately equal to the objects. Nonetheless, it is still difficult to locate the complex remote-sensing objects precisely solely based on the first regression stage. Thus, in the second regression stage of Gradual RPN, the focus shifts to predicting more precise candidates of complex remote-sensing objects from the coarse boxes generated in the previous stage. Due to the irregular shape of remote-sensing targets, the further feature extraction layer in the second stage utilizes a DCN [8] to extract the key features of target objects. Ultimately, the feature maps that have been enhanced with DCN will be supplied to a classifier and regressor, which will then generate the foreground probability and the precise location of each proposal.

The loss function of our Gradual RPN $\mathcal{L}_{\text{rpn}}$ is composed of four parts

$$\mathcal{L}_{\text{rpn}} = \mathcal{L}_{\text{reg}}^{s_1} + \mathcal{L}_{\text{reg}}^{s_2} + \mathcal{L}_{\text{cls}}^{s_2} + \mathcal{L}_{\text{offset}}^{s_2} \tag{6}$$

where the first two parts of $\mathcal{L}_{\text{rpn}}$ are the bounding box regression losses, the third part of $\mathcal{L}_{\text{rpn}}$ is the classification loss in the second stage, and the last part of $\mathcal{L}_{\text{rpn}}$ is a loss to penalize irrelevant feature points.

The regression losses in the first and second stages are both Intersection over Union (IoU) losses [51] and the detailed calculation of the IoU losses we adopted can be written as follows:

$$\mathcal{L}_{\text{reg}}^{s_i} = \frac{\lambda_{\text{reg}}}{N_{\text{pos}}^{s_i}} \sum_{j=0}^{N_{\text{pos}}^{s_i}} -\ln\left(\text{IoU}\left(B_j^{s_i}, B_j^{\text{gt}}\right)\right), \quad i \in \{1, 2\} \tag{7}$$

where $i, i \in \{1, 2\}$, is the ith regression stage, and $N_{\text{pos}}^{s_i}$ denotes the number of predicted boxes assigned as positive boxes in the ith regression stage. $B_j^{s_i}$ is the jth predicted box at stage i, which is a part of positive predicted boxes generated by Bi in Fig. 2(c). B_j^{gt} refers to the ground-truth box corresponding to the jth positive predicted box at stage i. The hyperparameter λ_{reg} serves as a balancing factor between the classification loss and regression losses, and we set it to 10 by default.

A cross-entropy loss is then utilized as the classification loss in the second stage of regression, and it can be denoted as

$$\mathcal{L}_{\text{cls}}^{s_2} = \frac{1}{N_{\text{sam}}} \sum_{i=0}^{N_{\text{sam}}} -(y_i \cdot \log(p_i) + (1 - y_i) \cdot \log(1 - p_i)) \tag{8}$$

where N_{sam} is the number of sampled boxes, p_i represents the predicted probability that the ith sampled box belongs to foreground, while y_i represents the ground truth label of the ith sampled box. These predicted probabilities are all predicted by C2 in Fig. 2(c).

In addition, to avoid the DCN extracting too many irrelevant features that are too far away, we designed a loss to penalize feature points that are beyond candidate boxes as the flowing shows

$$\mathcal{L}_{\text{offset}}^{s_2} = \frac{1}{N} \sum_{i=1}^{N} \mathbb{I}\left(\mathcal{O}_i^{s2}\right) \left\| \mathcal{O}_i^{s2} - \mathcal{C}_i^{\text{gt}} \right\|_2^2 \tag{9}$$

where $\mathcal{O}_i^{s2}$ is the offset point of the ith feature point in the DCN layer of Fig. 2(c), $\mathcal{C}_i^{\text{gt}}$ is the ground-truth box of the ith feature point, and $\mathbb{I}(\mathcal{O}_i^{s2})$ is an indicator function that is 1 if the ith feature point is inside the ground-truth box and 0 otherwise.

By the step-by-step regression process and enhanced feature extraction network, the Gradual RPN we proposed can generate more precise proposals in few-shot scenarios. But in the fine-tuning stage, more precise candidate boxes will reduce the diversity of proposals which will harm the performance of the following classification module. Consequently, we solve the proposal diversity problem by simply cropping the proposals in the fine-tuning stage.

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TABLE I
FOUR DIFFERENT BASE/NOVEL SPLIT SETTINGS ON THE DIOR DATASET

Split	Novel					Base
1	Baseball field	Basketball court	Bridge	Chimney	Ship	Rest
2	Airplane	Airport	Expressway toll station	Harbor	Ground track field	Rest
3	Dam	Golf course	Storage tank	Tennis court	Vehicle	Rest
4	Express service area	Overpass	Stadium	Train station	Windmill	Rest

TABLE II
FSOD PERFORMANCE ON THE DIOR TEST SET UNDER THREE, FIVE, TEN, AND 20-SHOT SETTINGS. THE BEST TWO PERFORMANCES ARE HIGHLIGHTED WITH RED AND BLUE, RESPECTIVELY. † REPRESENTS THE EXPERIMENTAL RESULTS IN [40]

Split	Method	Year	3-shot	5-shot	10-shot	20-shot
1	Meta R-CNN [35]	ICCV2019	12.02	13.09	14.07	14.45
	FsDetView [36]	ECCV2020	13.19	14.29	18.02	18.01
	TFA w/cos [3]	ICML2020	16.07	15.36	16.45	18.93
	P-CNN† [40]	TGRS2021	18.00	22.80	27.60	29.60
	FSOD [4]	CVPR2021	15.94	20.27	24.22	28.16
	FSCE [9]	CVPR2021	**27.91**	**28.60**	33.05	37.55
	MSOCL [45]	TGRS2022	24.97	27.27	**33.37**	**39.22**
	ICPE [10]	AAAI2023	11.68	12.34	12.95	14.33
	VFA [11]	AAAI2023	21.94	21.27	23.32	24.28
	SAE-FSDet (Ours)		**28.80**	**32.40**	**37.09**	**42.46**
2	Meta R-CNN [35]	ICCV2019	8.84	10.88	14.90	16.71
	FsDetView [36]	ECCV2020	10.83	9.63	13.57	14.76
	TFA w/cos [3]	ICML2020	6.81	7.53	8.93	11.05
	P-CNN† [40]	TGRS2021	**14.50**	**14.90**	**18.90**	**22.80**
	FSOD [4]	CVPR2021	9.35	9.73	14.84	16.20
	FSCE [9]	CVPR2021	13.17	14.07	15.79	20.93
	MSOCL [45]	TGRS2022	13.31	13.40	15.00	18.15
	ICPE [10]	AAAI2023	10.92	10.56	12.39	13.18
	VFA [11]	AAAI2023	12.10	12.70	14.72	15.47
	SAE-FSDet (Ours)		**13.99**	**15.65**	**17.41**	**21.34**
3	Meta R-CNN [35]	ICCV2019	9.10	12.29	11.96	16.14
	FsDetView [36]	ECCV2020	7.49	12.61	11.49	17.02
	TFA w/cos [3]	ICML2020	8.73	9.31	12.19	16.97
	P-CNN† [40]	TGRS2021	**16.50**	**18.80**	23.30	28.80
	FSOD [4]	CVPR2021	10.40	10.74	12.26	11.52
	FSCE [9]	CVPR2021	15.59	16.24	**23.75**	**28.89**
	MSOCL [45]	TGRS2022	13.11	15.07	23.39	27.44
	ICPE [10]	AAAI2023	10.56	11.21	12.38	13.08
	VFA [11]	AAAI2023	11.97	13.19	15.45	17.61
	SAE-FSDet (Ours)		**16.74**	**19.07**	**28.44**	**29.88**
4	Meta R-CNN [35]	ICCV2019	13.94	15.84	15.07	18.17
	FsDetView [36]	ECCV2020	14.28	15.95	15.37	16.96
	TFA w/cos [3]	ICML2020	9.54	13.82	13.82	16.61
	P-CNN† [40]	TGRS2021	15.20	17.50	18.90	25.70
	FSOD [4]	CVPR2021	11.84	12.98	17.17	18.46
	FSCE [9]	CVPR2021	**17.45**	**20.42**	**22.22**	**24.96**
	MSOCL [45]	TGRS2022	10.40	12.29	16.64	22.67
	ICPE [10]	AAAI2023	14.45	14.52	15.95	15.61
	VFA [11]	AAAI2023	15.52	17.76	18.62	20.05
	SAE-FSDet (Ours)		**17.27**	**20.48**	**22.69**	**26.75**

IV. EXPERIMENTS AND RESULTS

In this section, we evaluate the performance of our proposed method on two widely used public RSI datasets. To validate the effectiveness of our proposed method, we compare SAE-FSDet with both universal few-shot detectors and remote-sensing few-shot detectors and conduct extensive ablation studies.

A. Datasets

The DIOR dataset is a large-scale optical remote-sensing benchmark for object detection released by [6]. Images in the DIOR dataset are all collected from Google Earth and the dataset contains 23 463 images and 192 472 instances of 20 classes. The object classes in the DIOR dataset include *airplane, airport, baseball field, basketball court, bridge,*

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TABLE III
FSOD PERFORMANCE ON THE NWPU VHR-10 DATASET UNDER THREE, FIVE, TEN, AND 20-SHOT SETTINGS

Method	3-shot	5-shot	10-shot	20-shot
Meta-RCNN [35]	20.51	21.77	26.98	28.24
FsDetView [36]	24.56	29.55	31.77	32.73
TFA w/cos [3]	16.17	20.49	21.22	21.57
P-CNN [40]	**41.80**	**49.17**	**63.29**	66.83
FSOD [4]	10.95	15.13	16.23	17.11
FSCE [9]	41.63	48.80	59.97	**79.60**
ICPE [10]	6.10	9.10	12.00	12.20
VFA [11]	13.14	15.08	13.89	20.18
SAE-FSDet (Ours)	**57.96**	**59.40**	**71.02**	**85.08**

chimney, dam, expressway service area, expressway toll station, harbor, golf course, ground track field, overpass, ship, stadium, storage tank, tennis court, train station, vehicle, and *windmill.* The entire dataset is divided into three parts: training set, validation set, and testing set, containing 5682, 5863, and 11 738 images, respectively. The spatial resolution of images in the DIOR dataset ranges from 0.5 to 30 m and the size of all images is 800 × 800 pixels.

NWPU VHR-10 is a very high-resolution (VHR) optical remote-sensing dataset released by Cheng et al. [52]. The NWPU VHR-10 dataset contains 800 optical images collected from Google Earth and the ISPRS Vaihingen dataset [12]. All the 800 images in the NWPU VHR-10 are composed of 150 "negative samples" without any annotated objects and 650 "positive samples" with at least one annotated object. The dataset contains ten geospatial object classes in total, including *airplane, baseball diamond, basketball, bridge, court, ground track field, harbor, ship, storage tank, tennis court,* and *vehicle.*

B. Experimental Settings and Evaluation Metrics

To evaluate the detection performance of our proposed method under the FSOD setting, we divide the classes of each dataset into two parts: base classes and novel classes. Each class of the base classes contains sufficient annotated samples and each class of the novel classes only contains K instances. For a more accurate evaluation of performance, we select $K = 3, 5, 10$, and 20 annotations for each class from the training and validation sets in the K-shot detection task.

Specifically, for the DIOR dataset, we follow the FSOD setting in [40] and adopt four different novel/base splits, each of which includes 15 base classes and five novel classes, detailed in Table I. For the NWPU VHR-10 dataset, to compare with other methods fairly, we choose to follow the class split introduced in [53] and we unbiasedly select *airplane, baseball diamond*, and *tennis court* as novel classes. The rest classes in the NWPU VHR-10 dataset are referred to as base classes.

For evaluation metrics, we use mean average precision (mAP) to evaluate the performance of detectors. We follow the PASCAL VOC2007 benchmark [7] to calculate the mAP, which takes the average of 11 precision values when recall increases from 0 to 1 with a step of 0.1 and an IoU threshold of 0.5.

C. Implementation Details

Our proposed method is based on TFA [3] and we adopt the pretrained ResNet101 [54] with FPN [55] as the backbone. In the base training stage, the model is trained by 18 epochs, where the learning rate starts from 0.01 and decreases to 0.001 at 12 epochs and 0.0001 at 16 epochs. In the fine-tuning stage, the model is fine-tuned by 108 epochs with a learning rate of 0.001. In both the base training and the fine-tuning stage, we use stochastic gradient descent (SGD) as a gradient backward method with a momentum of 0.9 and a weight decay coefficient of $1e^{-4}$. The training batch size is set to 2 on one NVIDIA GeForce RTX 4090 GPU. During the inference phase, the confidence threshold and IoU threshold in non maximum suppression (NMS) are set to 0.05 and 0.5, respectively. Besides, any additional data augmentation strategies are not utilized in both the training and inference phases. We run each experiment of our proposed method three times and report the average results over three repeated experiments.

D. Experimental Results and Comparisons

1) Results on the DIOR Dataset: To verify the effectiveness of our proposed SAE-FSDet, we make comprehensive comparisons with several state-of-the-art FSOD methods, including Meta R-CNN [35], FsDetView [36], TF-A [3], P-CNN [40], FSOD [4], FSCE [9], MSOCL [45], ICPE [10], and VFA [11]. These comparison methods include FSOD methods designed for natural images [3], [9], [10], [11], [35], [36] and FSOD methods specifically designed for RSIs [40], [45]. The FSOD methods designed based on natural scenes encompass both meta-learning-based methods [10], [11], [35], [36] and fine-tune-based methods [3], [9]. The results of all comparison methods are shown in Table II.

As Table II shows, almost all FSOD approaches based on natural scenes suffer from severe performance degradation in the remote-sensing FSOD task and the performance degradation proves that FSOD methods based on natural scenes are not fully applicable to remote-sensing scenes. Meanwhile, our proposed SAE-FSDet achieves state-of-the-art performance in most few-shot splits and attains top-2 detection performance across all compared methods. It shows the effectiveness of our proposed SAE-FSDet. Table II also shows that P-CNN [40] exhibits superior performance compared to other methods in some experiments of the second class split. We argue that novel classes in the second class split contain more thin instances, and the strong rotation augmentation utilized in P-CNN [40] enhances the ability to detect thin instances. Nevertheless, since P-CNN [40] does not consider the widespread inconsistent label assignment in RSIs, the performance of P-CNN still falls behind that of SAE-FSDet in most few-shot cases, especially in the first split.

In addition to qualitative analysis, we also visualize some detection results and failure cases in Figs. 4 and 5. As Fig. 4 shows, our proposed SAE-FSDet can detect both base and novel objects with just ten samples. Furthermore, we show some typical failure cases in Fig. 5. The reasons for failure in these cases can be roughly divided into three types: 1) large interclass similarity; 2) dense object distribution; and 3) small

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TABLE IV
ABLATION STUDY OF EACH PROPOSED MODULE IN OUR SAE-FSDET ON THE FIRST SPLIT OF THE DIOR DATASET AND THE NWPU VHR-10 DATASET

Method	Componets		Shots								Params	FLOPs	FPS
	Gradual RPN	LCC	DIOR				NWPU VHR-10						
			3	5	10	20	3	5	10	20			
TFA w/cos			16.07	15.36	16.45	18.93	16.17	20.49	21.22	21.57	60.25	178.97	54.3
TFA† w/cos			21.44	27.64	32.70	37.81	42.60	49.27	58.39	76.87			
SAE-FSDet	✓		22.47	27.91	33.43	39.75	50.31	56.61	62.61	82.80	60.89	181.13	45.7
SAE-FSDet	✓	✓	28.80	32.40	37.09	42.46	57.96	59.40	71.02	85.08	60.89	181.13	45.6



Fig. 4. Visualization of the detection results on the DIOR test set in the first base/novel split under a ten-shot setting.



Fig. 5. Visualization of failure cases on the DIOR test set in the first base/novel split under a ten-shot setting. 1) Some bridges are misclassified into overpasses and harbors. 2) Dense ships are difficult to detect. 3) Small ships that lack discriminatory features are also difficult to detect.

TABLE V
MAP COMPARISONS BETWEEN THE CONVENTIONAL CLASSIFIER AND OUR PROPOSED LCC ON THE DIOR TEST SET IN THE FIRST BASE/NOVEL SPLIT

Method	Shot			
	3	5	10	20
TFA w/cos	20.22	20.96	27.42	35.30
TFA† w/cos	32.61	37.27	44.76	48.77
LCC	40.40	49.45	56.21	63.42

TABLE VI
AVERAGE RECALL COMPARISONS BETWEEN THE CONVENTIONAL RPN AND OUR PROPOSED GRADUAL RPN ON THE DIOR TEST SET IN THE FIRST BASE/NOVEL SPLIT

Method	Shot			
	3	5	10	20
RPN(Frozen)	20.98	20.98	20.98	20.98
RPN(Unfrozen)	28.96	33.16	34.20	38.38
Gradual RPN	41.40	42.60	46.56	50.24

objects. Since the model is trained by many base examples, some novel objects are misclassified into base classes that are similar to them. Additionally, the dense distribution of objects presents a challenge in accurately detecting them. Small objects, such as ships, also lack deterministic features, making them difficult to detect.

2) Results on the NWPU VHR-10 Dataset: We also conduct comparison experiments on the NWPU VHR-10 dataset [52] and compare our proposed method with several FSOD methods, including Meta-RCNN [35], FsDetView [36], TFA [3], P-CNN [40], FSOD [4], FSCE [9], ICPE [10], and VFA [11]. The results of all comparison methods are shown in Table III.

We can observe that our proposed SAE-FSDet outperforms all competing methods, including both meta-learning-based methods and fine-tune-based methods in all shots, by solving the inconsistent label assignment of novel instances and enhancing the recall of RPN. With only 20 labeled instances, our SAE-FSDet can achieve 85.1% mAP, which is comparable to the performance of a detector trained from a large dataset. Lastly, the detection results under the ten-shot setting in the first base/novel split are shown in Fig. 6.

E. Ablation Studies

Extensive ablation studies are conducted in this section to better understand the effectiveness of each proposed module in our SAE-FSDet. All experiments are performed on the first base/novel split of the DIOR test set or the NWPU VHR-10 test set.

1) Ablation Study for the Components of Our Proposed Method: To analyze the importance of each module we proposed and verify the robustness of SAE-FSDet, we per-

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TABLE VII

ABLATION STUDY FOR FEATURE EXTRACTORS ON THE DIOR TEST SET IN THE FIRST BASE/NOVEL SPLIT. Δ REPRESENTS THE IMPROVEMENT ACHIEVED BETWEEN THE BASELINE AND OUR PROPOSED METHOD

Method	Backbone	Shot							
		3-shot	Δ	5-shot	Δ	10-shot	Δ	20-shot	Δ
TFA† w/cos	ResNet-101	21.44		27.64		32.70		37.81	
SAE-FSDet	ResNet-101	27.07	↑5.63	32.40	↑4.76	37.09	↑4.39	42.46	↑4.65
TFA† w/cos	ViT	14.80		18.96		25.58		31.86	
SAE-FSDet	ViT	19.10	↑4.30	21.36	↑2.40	26.32	↑0.74	33.30	↑1.44
TFA† w/cos	ViTAE-rvsa	13.07		15.17		25.46		34.10	
SAE-FSDet	ViTAE-rvsa	23.23	↑10.16	25.51	↑10.34	30.85	↑5.39	39.53	↑5.43

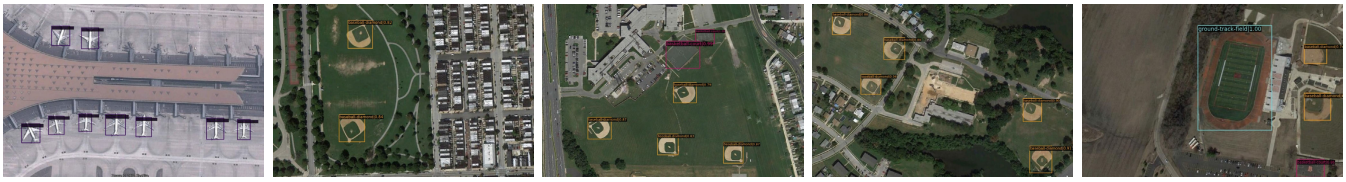


Fig. 6. Visualization of the detection results on NWPU VHR-10 in the first base/novel split under a ten-shot setting.

form ablation studies on both the DIOR dataset and the NWPU VHR-10 dataset. The detailed results are presented in Table IV.

At first, we set TFA [3], a simple yet effective few-shot object detector for natural images, as our baseline model. However, TFA [3] is designed for natural images and does not take the complexity of remote-sensing targets into account. To address this limitation, we unfreeze the RPN head and RoI feature extractor while keeping the backbone frozen. By doing this, the detector can better adapt to new classes that exhibit significant differences from base classes. The improved method of TFA [3] is denoted as TFA† and the performance of TFA† is demonstrated in the second row of Table IV.

Next, we employ our designed Gradual RPN as a replacement for the conventional RPN, to generate more accurate proposals. As demonstrated in the third row of Table IV, with the addition of Gradual RPN, the mAP score compared to the baseline improved by 3.65% on average. It indicates that proposals with higher quality can assist the detector in achieving better detection performance. We also note that the performance improvement of Gradual RPN on the NWPU VHR-10 dataset is more significant than on the DIOR dataset. We argue that the classification difficulty of DIOR is higher than that of the NWPU VHR-10 dataset, and the lower classification accuracy of the detector on the DIOR dataset limits the performance improvement of Gradual RPN. A more comprehensive analysis of Gradual RPN will be presented in Section IV-E3.

Then, the LCC is used to replace the conventional classifier in the fine-tuning stage. As shown in the fourth row of Table IV, the LCC further boosts the performance by 6.33%/4.49%/3.66%/2.71% under the three-shot/five-shot/ten-shot/20-shot setting on DIOR dataset and 7.65%/2.79%/8.41%/2.28% under the three-shot/five-shot/ten-shot/20-shot setting on NWPU VHR-10 dataset. We observe that by avoiding the inconsistent label assignment of novel instances, LCC can significantly improve the performance of the detector in remote-sensing few-shot scenarios.

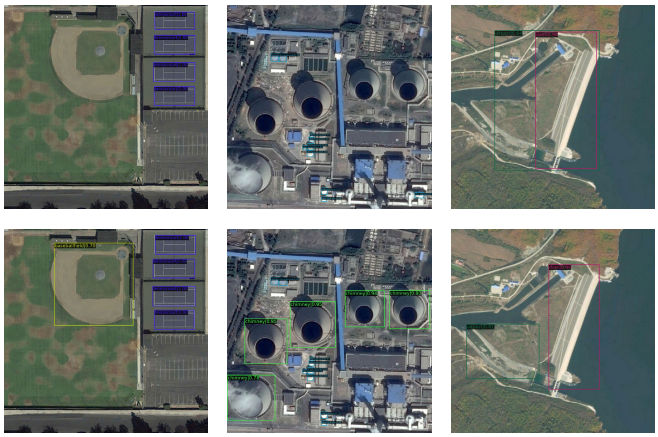


Fig. 7. Visualization of the compared results between the baseline (first row) and SAE-FSDet (second row) on the DIOR test set in the first base/novel split under a ten-shot setting.

In Table IV, we also provide the number of parameters for each module as well as the inference speed. Since Gradual RPN involves gradual regression on proposals, the inference speed of the detector may experience a slight decrease. However, it still achieves a frame rate of 45.7 frames/s on a single RTX 4090 device, while booting the performance by an average of 3.65%. Besides that, as the hierarchical computation in LCC is only utilized for classification scores, the additional computation is small and can be considered negligible compared to the overall network. Consequently, the addition of LCC has little effect on the ultimate inference speed.

At last, we visualize some comparative results between the baseline and our proposed SAE-FSDet. As shown in Fig. 7, compared to the baseline detector, our proposed SAE-FSDet can detect more novel instances with only a few samples benefiting from our proposed Gradual RPN and LCC.

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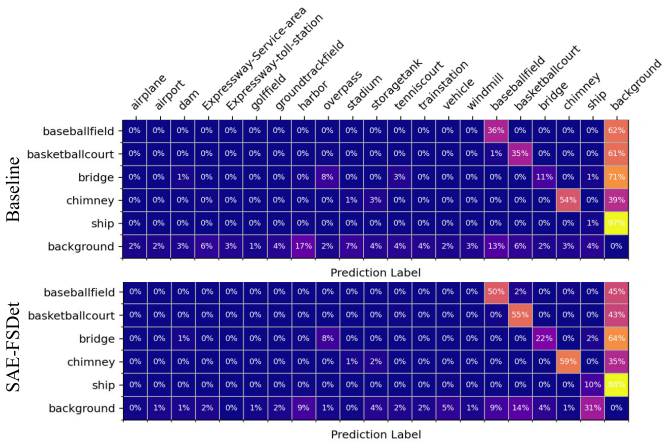


Fig. 8. Classification confusion matrix of our baseline (top) and SAE-FSDet (bottom) on the DIOR test set under the first base/novel split under the ten-shot setting. In the baseline method, a large number of novel objects are misclassified as background due to inconsistent label assignment. However, with the assistance of LCC, the misclassification in SAE-FSDet significantly decreases.

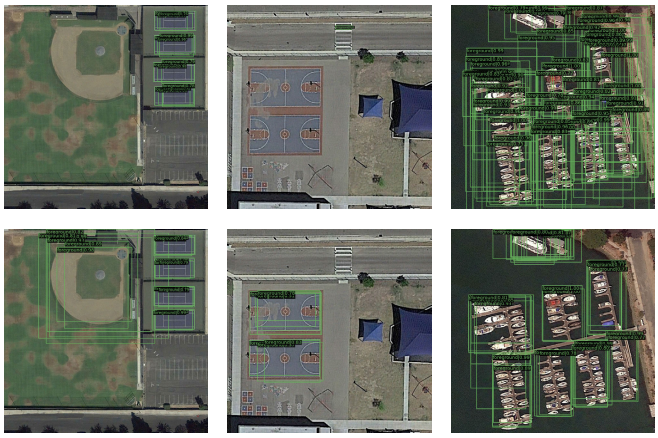


Fig. 9. Visualization of the compared results between the simple RPN (first row) and our proposed Gradual RPN (second row) on the DIOR test set in the first base/novel split under a ten-shot setting.

2) Ablation for the Performance of LCC: LCC is proposed to avoid the inconsistent assignment of unseen novel objects between the base training stage and the fine-tuning stage. Moreover, the LCC can also leverage the unlabeled novel samples in the base dataset with sufficient samples to improve the detection performance of novel classes without any additional effort. For a fairer comparison and to remove the inference of RPN on the classifier, all the following experiments directly use the ground truth boxes as proposals to be classified. Table V shows the comparison results between the classification performance of the baseline classifier and our proposed LCC. We can observe that insufficient feature extraction of the RoI feature extractor limits the classification accuracy and unfreezing the RoI feature extractor in the fine-tuning stage can improve the classification accuracy of novel classes. Additionally, compared with TFA†, the detector with our proposed LCC further improves the average performance by 11.5%.

In addition to quantitative analysis, the confusion matrix of novel classes is visualized in Fig. 8 to better understand the effectiveness of our proposed LCC. The figure shows the classification confusion matrix of the novel classes under the ten-shot setting between the baseline TFA† and our proposed LCC. As illustrated in the confusion matrix, in the baseline method, a large number of novel objects are classified as background categories due to the problem of inconsistent label assignment. However, with the assistance of LCC, the misclassification of novel instances into backgrounds has significantly reduced.

3) Ablation for the Performance of Gradual RPN: The Gradual RPN we proposed aims to detect complex remote-sensing objects by two regression steps in few-shot scenarios. To verify the effectiveness of our proposed Gradual RPN, we compare the overall recall of the novel classes for different approaches, as Table VI shows. We observe that unfreezing RPN in the fine-tuning phase improves the low recall of RPN, but the recall of unfrozen RPN is still unsatisfactory. Besides, with a more delicate design, our proposed Gradual RPN can significantly improve the average recall of novel classes and generate more high-quality proposals by 11.5%. The results show that Gradual RPN can generate more high-quality proposals.

Besides the above quantitative analysis, we also visualize the comparative results in Fig. 9. We can observe that, compared to the simple RPN, Gradual RPN captures more proposals, and the proposals generated from Gradual RPN are more accurate benefiting from the step-by-step regression process and enhanced feature extraction in Gradual RPN.

4) Ablation of Feature Extractors: Currently, some transformer-based feature extractors demonstrate superior robustness compared to CNN-based feature extractors in both natural images [56], [57], [58] and RSIs [59], [60]. Fortunately, thanks to the flexible design of Gradual RPN and LCC, our proposed SAE-FSDet is compatible with both CNN-based feature extractors and transformer-based extractors. To better verify the robustness of our proposed method, we conduct ablation studies on feature extractors. Without sacrificing generalization, we choose one CNN-based feature extractor ResNet-101 [54], and two transformer-based feature extractors ViT [61] and ViTAE-rvsa [60] as our base feature extractors. The ResNet-101 is supervised and pretrained on ImageNet [62]. The ViT and ViTAE-rvsa utilize MAE [61] for unsupervised pretraining on ImageNet and MillionAID [63] datasets, respectively. The results demonstrate that our proposed SAE-FSDet works well on both CNN-based and transformer-based feature extractors.

The experimental results in Table VII demonstrate the efficacy of our proposed method in both transformer-based and CNN-based feature extractors. Particularly in experiments with quite a few training samples, the performance improvement of our proposed method is substantial. For example, in three-shot experiments, SAE-FSDet significantly improves performance by 6.70% on average. While the problems mentioned in Section I become less prominent as the number of shots increases, the SAE-FSDet method, designed to address these issues, still achieves at least a 3.51% performance improvement on average.

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It is also evident that CNN-based feature extractors outperform transformer-based feature extractors in the FSOD of RSIs. This superior performance can be attributed to the tendency of transformer-based feature extractors to overfit in the case of a few training samples. We argue that much of the robustness of transformer-based feature extractors comes from the large number of training parameters and the diversity of training data. Therefore, in the absence of a large amount of training data, transformer-based feature extractors are more prone to overfitting compared to CNN-based feature extractors.

V. Conclusion

In this article, we focus on solving the inconsistent label assignment of novel instances in the base dataset as well as the low recall issue of RPN in current remote-sensing FSOD methods. To do so, we introduce a sample-efficient few-shot detector, named SAE-FSDet, which is based on a transfer-learning framework. Specifically, SAE-FSDet first employs an LCC to solve the label inconsistency problem in existing FSOD methods. The LCC can further improve the detection performance for novel classes by leveraging unlabeled instances. In addition, a stepwise gradual position regressor named Gradual RPN is proposed to utilize the characteristics of complex remote-sensing objects and generate adequate high-quality proposals. The experimental results show that SAE-FSDet can improve the mAP of novel classes by up to 8.4% and Gradual RPN can improve the recall of RPN by up to 12.3%.

The number of available samples is the key that limits the performance in few-shot learning. Therefore, to alleviate the data scarcity of FSOD, data augmentation is a promising direction for few-shot learning. In the future, we will use some novel data generation methods such as DDPM [64], GLIDE [65], and stable-diffusion [66] to alleviate the difficulty of remote-sensing FSOD.

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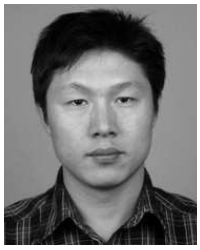
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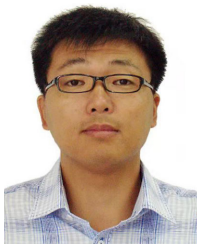
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